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 Studierichting: Informatica
 Oefeningenreeks 3H nr. 11.3

$$r^2 = 9 \sin 2\theta \Leftrightarrow r = \sqrt{9 \sin 2\theta}$$

1. Domein

$$p = \frac{2\pi}{2} = \pi$$

$\Rightarrow$ we onderzoeken op het beperkt domein $[0, \pi]$

Binnen dit interval geldt: (Omdat in $\mathbb{R}$ de wortel van een negatief getal niet bestaat)

$$\sin 2\theta \geq 0 \iff \theta \in \left[0, \frac{\pi}{2}\right]$$

$$\Rightarrow \text{dom } r = \left[0, \frac{\pi}{2}\right]$$

2. Tekenonderzoek van $r(\theta)$

$$r^2 = 9 \sin 2\theta$$

$$\sin 2\theta = 0$$

$$2\theta = k\pi$$

$$\theta = \frac{k\pi}{2}$$

$$\Rightarrow r^{(-1)}(0) = \left\{0, \frac{\pi}{2}\right\}$$

θ	0	$\frac{\pi}{2}$
$r(\theta)$	0	0
	$+$	

3. Tekenonderzoek van $r'(\theta)$

$$r = \sqrt{9 \sin 2\theta}$$

$$r = 3\sqrt{\sin 2\theta}$$

$$r(\theta) = 3(\sin 2\theta)^{\frac{1}{2}}$$

$$r'(\theta) = 3 \cdot \frac{1}{2} (\sin 2\theta)^{-\frac{1}{2}} \cdot 2 \cos 2\theta$$

$$r'(\theta) = \frac{3}{\sqrt{\sin 2\theta}} \cos 2\theta$$

$$\cos 2\theta = 0$$

$$2\theta = \frac{\pi}{2} + k\pi$$

$$\theta = \frac{\pi}{4} + \frac{k\pi}{2}$$

$$\Rightarrow r'^{(-1)}(0) = \left\{ \frac{\pi}{4} \right\}$$

$$\frac{\theta}{r'(\theta)} \left| \begin{array}{c} \frac{\pi}{4} \\ 0 \end{array} \right. \begin{array}{c} + \\ - \end{array}$$

Maximum:

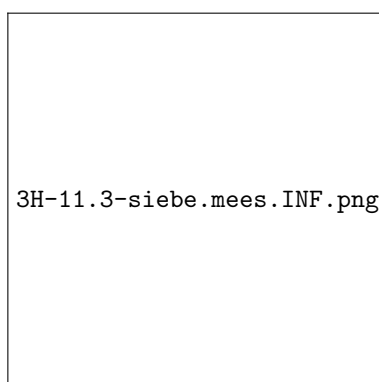
$$r\left(\frac{\pi}{4}\right) = \sqrt{9 \sin\left(2 \cdot \frac{\pi}{4}\right)}$$

$$r\left(\frac{\pi}{4}\right) = \sqrt{9 \sin \frac{\pi}{2}}$$

$$r\left(\frac{\pi}{4}\right) = \sqrt{9}$$

$$\Rightarrow \max r\left(\frac{\pi}{4}\right) = 3$$

4. Schets



$\Rightarrow$ Samenvattende tabel

θ	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$
$r(\theta)$	0	+	0
$r'(\theta)$	+	0	-
$r(\theta)$	$\nearrow$	$M(3)$	$\searrow$

References